

DevOps Project Documentation:

Project Title:

End-to-End CI/CD Pipeline with Kubernetes Monitoring on AWS

Project Description:

Designed and implemented a complete DevOps pipeline to automate application deployment using Jenkins, Docker, and Kubernetes on AWS EC2. Integrated Prometheus and Grafana for real-time monitoring and visualization of system and application metrics.

Architecture Overview:

- Source Code → GitHub
 - CI/CD → Jenkins
 - Containerization → Docker
 - Orchestration → Kubernetes (K8s)
 - Monitoring → Prometheus
 - Visualization → Grafana
 - Infrastructure → AWS EC2 (Ubuntu)
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Tools & Technologies Used:

- **Cloud:** AWS EC2
- **CI/CD:** Jenkins
- **Containerization:** Docker
- **Orchestration:** Kubernetes (kubectl)
- **Monitoring:** Prometheus
- **Visualization:** Grafana
- **Version Control:** Git & GitHub
- **OS:** Ubuntu Linux

Implementation Steps:

1. Infrastructure Setup

- Launched an AWS EC2 instance
- Installed Docker, Kubernetes (kubectl), Jenkins, and Helm

2. CI/CD Pipeline (Jenkins)

- Configured Jenkins on EC2
- Created pipeline job connected to GitHub repository
- Automated:
 - Pulling source code
 - Building Docker image
 - Running container

3. Containerization (Docker)

- Created Dockerfile for application
- Built Docker image (devops-app)
- Ran container using Jenkins pipeline

4. Kubernetes Deployment

- Created Kubernetes manifests:
 - deployment.yaml
 - service.yaml
- Deployed application using:

kubectl apply -f k8s/

- Exposed app using **NodePort service**

5. Monitoring Setup (Prometheus + Grafana)

- Installed using Helm:

helm install monitoring prometheus-community/kube-prometheus-stack

- Verified pods and services
- Accessed Grafana dashboard via NodePort

6. Dashboard & Metrics

- Connected Grafana with Prometheus
- Verified metrics using query:

up

- Created dashboards for:
 - CPU usage
 - Memory usage
 - Pod status

Key Features:

- Automated CI/CD pipeline
- Containerized application deployment
- Kubernetes-based scaling and orchestration
- Real-time monitoring with Prometheus
- Visual dashboards using Grafana
- Cloud-based deployment on AWS

Output / Result:

- Application successfully deployed on Kubernetes

- Accessible via EC2 public IP and NodePort
 - Monitoring dashboards displaying live metrics
 - All services (pods) running and healthy
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Challenges Faced:

- ImagePullBackOff issue (fixed using imagePullPolicy: Never)
- Jenkins workspace path errors
- Kubernetes service access (NodePort configuration)
- Grafana dashboard showing N/A (resolved using correct PromQL queries)

Pipeline script:

apiVersion: apps/v1

kind: Deployment

metadata:

name: devops-app

spec:

replicas: 2

selector:

matchLabels:

app: devops-app

template:

metadata:

labels:

app: devops-app

spec:

containers:

- name: devops-app

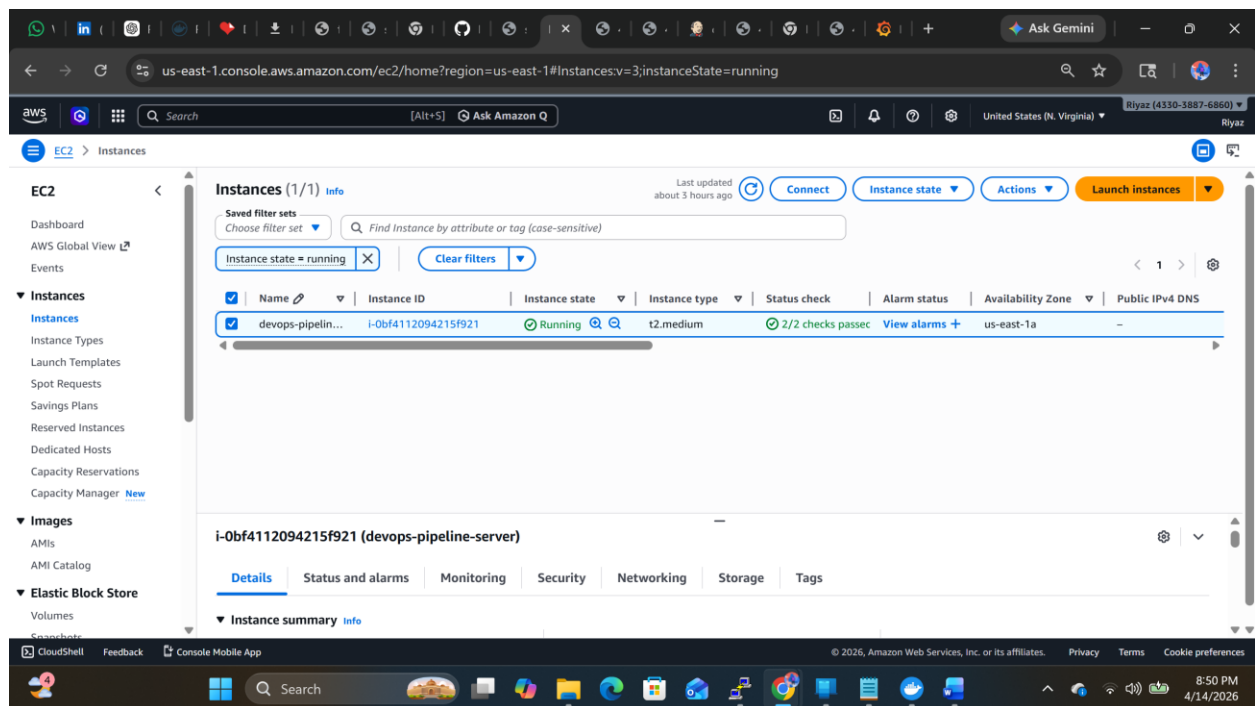
image: devops-app

imagePullPolicy: Never

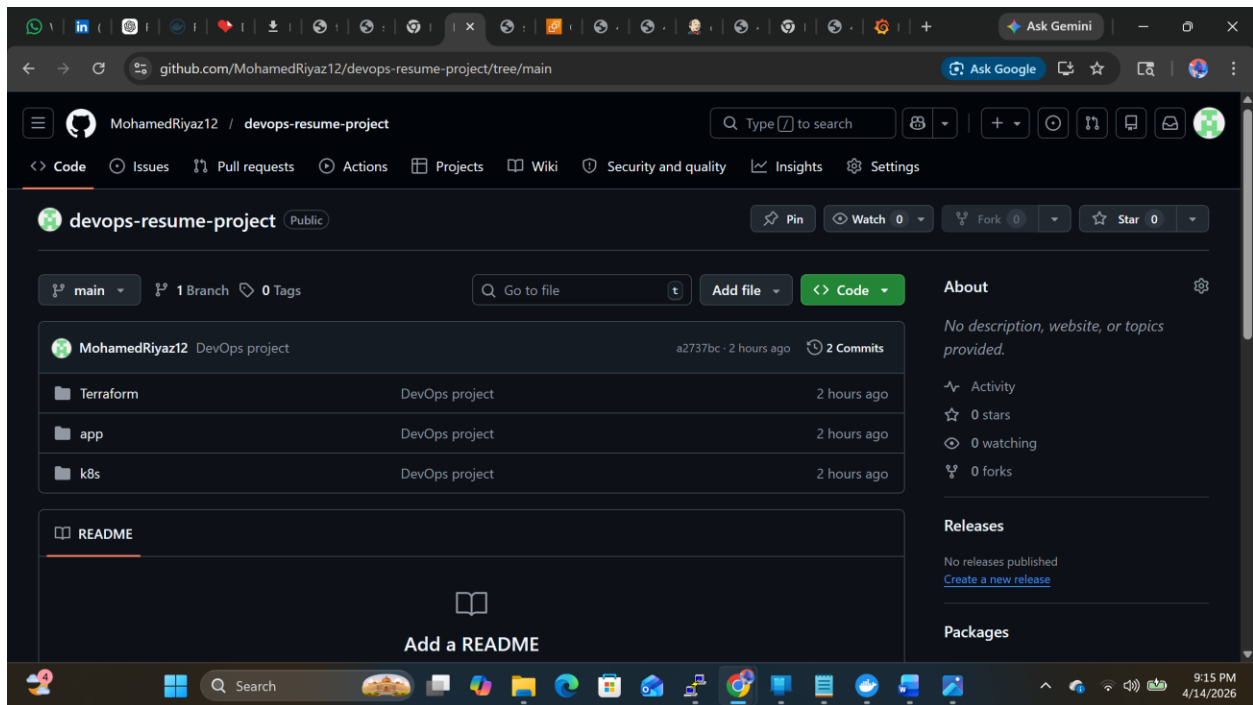
ports:

- containerPort: 5000

EC2 instance created:



Devops project github repository with required files:



Docker image created and docker container run:

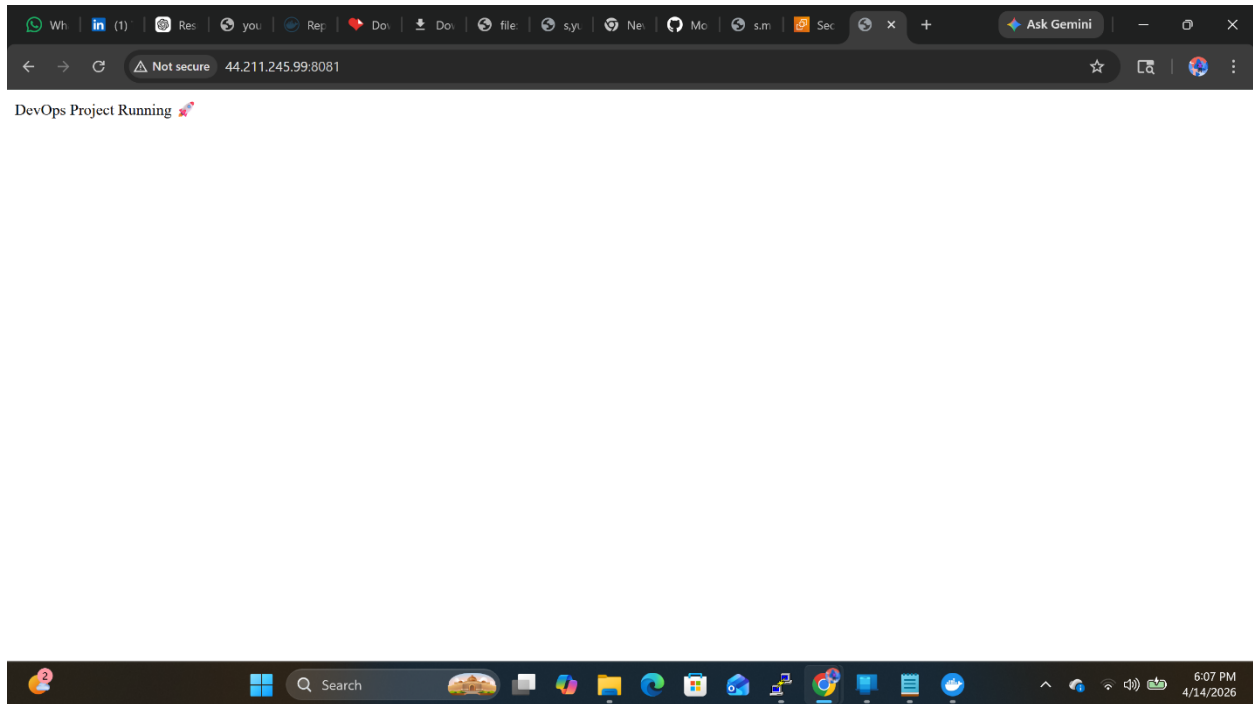
```

ubuntu@ip-172-31-14-240: ~/devops-project/app
----- 103.4/103.4 kB 9.9 MB/s eta 0:00:00
Collecting importlib-metadata>=3.6.0
  Downloading importlib_metadata-3.6.0-py3-none-any.whl (27 kB)
Collecting Jinja2>=3.1.2
  Downloading Jinja2-3.1.2-py3-none-any.whl (134 kB)
----- 134.9/134.9 kB 15.4 MB/s eta 0:00:00
Collecting blinker>=1.9.0
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting click>=8.1.3
  Downloading click-8.1.3-py3-none-any.whl (98 kB)
----- 98.2/98.2 kB 16.5 MB/s eta 0:00:00
Collecting itsdangerous>=2.2.0
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting Werkzeug>=3.1.0
  Downloading Werkzeug-3.1.0-py3-none-any.whl (226 kB)
----- 226.5/226.5 kB 31.5 MB/s eta 0:00:00
Collecting MarkupSafe>=2.1.1
  Downloading MarkupSafe-2.1.1-cp39-cp39-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
Collecting zipp>=3.20
  Downloading zipp-3.20.1-py3-none-any.whl (10 kB)
Installing collected packages: zipp, MarkupSafe, itsdangerous, click, blinker, Werkzeug, Jinja2, importlib-metadata, flask
Successfully installed blinker-1.9.0 click-8.1.3 flask-3.1.3 importlib-metadata-8.7.1 itsdangerous-2.2.0 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.8 zipp-3.20.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

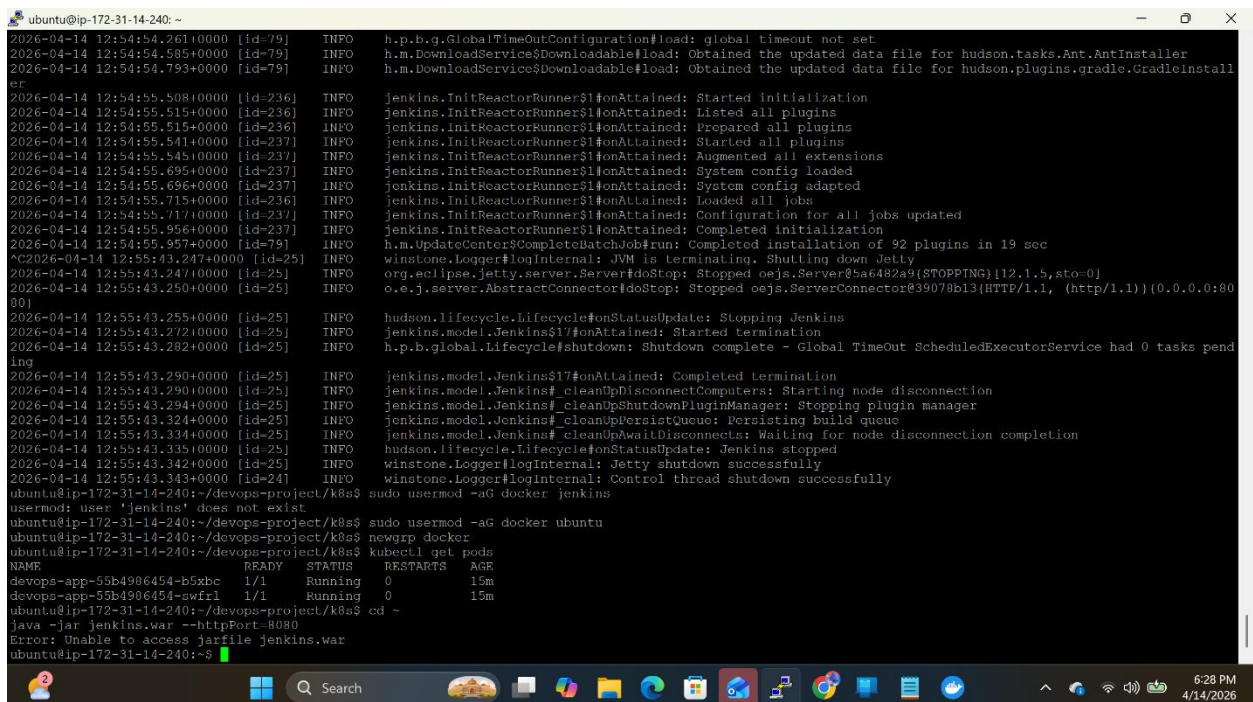
[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
----> Removed intermediate container 6d355e0ef4d5
----> 08046b1015a7
Step 5/5 : CMD ["python", "app.py"]
----> Running in 830034982ed7
----> Removed intermediate container 830034982ed7
----> d175ac445930
Successfully built d175ac445930
Successfully tagged devops-app:latest
ubuntu@ip-172-31-14-240:~/devops-project/app$ docker run -d -p 80:5000 devops-app
8238c3f42a8b273a706aa85c49aE80b30da574093525f954f75e984b9aa0
ubuntu@ip-172-31-14-240:~/devops-project/app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                               NAMES
8238c3f42a8b   devops-app    "python app.py"         About a minute ago   Up About a minute   0.0.0.0:80->5000/tcp, [::]:80->5000/tcp   competent_vaughan
ubuntu@ip-172-31-14-240:~/devops-project/app$ curl localhost
DevOps Project Running
ubuntu@ip-172-31-14-240:~/devops-project/app$

```

Initial output:



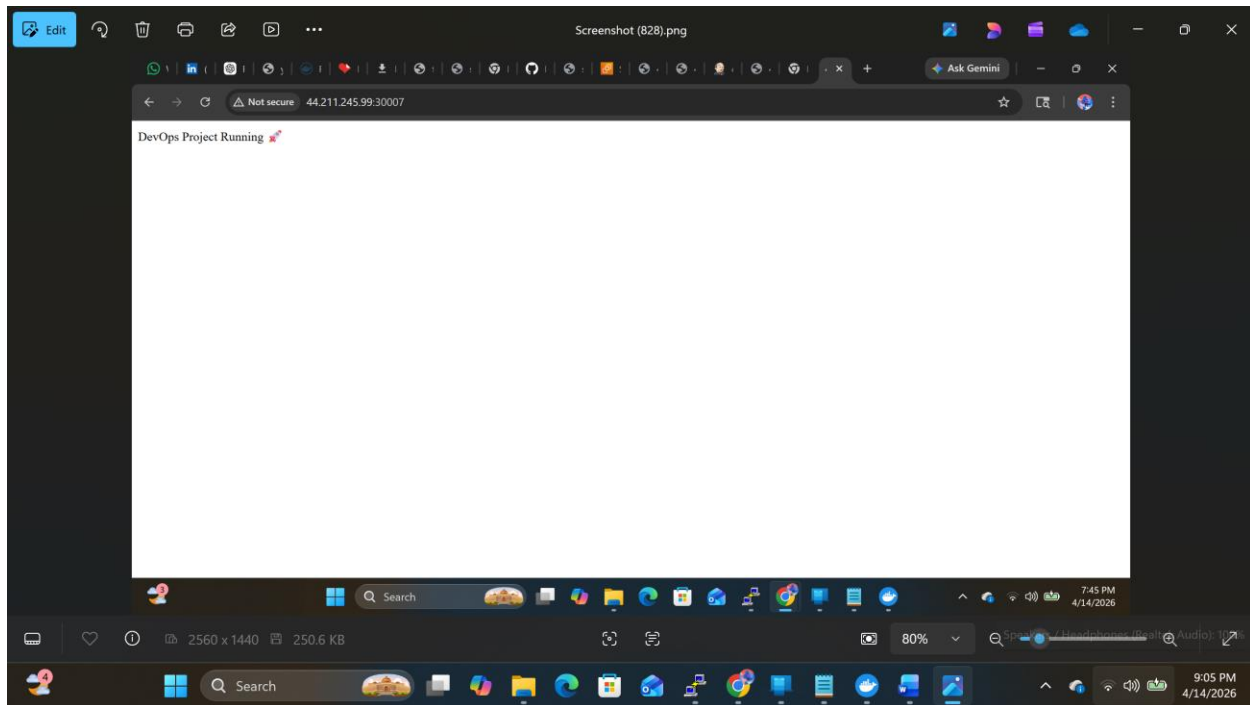
Jenkins installed:



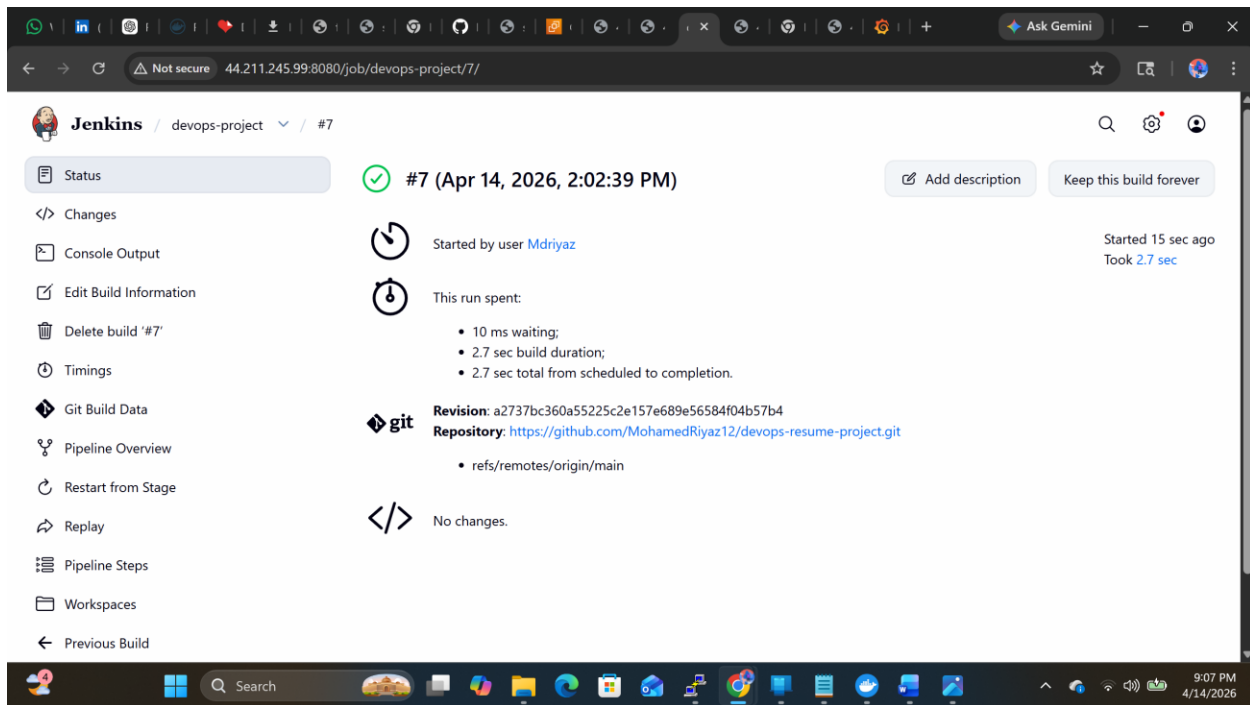
App deployed and service in Kubernetes:

```
ubuntu@ip-172-31-14-240: ~/devops-project/devops-resume-project
NAME                READY   STATUS    RESTARTS   AGE
devops-app-55b4986454-gn9ws   1/1     Running   0           7s
devops-app-55b4986454-gqkq2   1/1     Running   0           8s
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ kubectl get pods
kubectl get svc
NAME                READY   STATUS    RESTARTS   AGE
devops-app-55b4986454-gn9ws   1/1     Running   0           4m46s
devops-app-55b4986454-gqkq2   1/1     Running   0           4m47s
devops-app-848bd76fb9-x/gw/    0/1     ImagePullBackOff    0           51s
NAME                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
devops-service      NodePort      10.43.19.120 <none>        80:30007/TCP     12m
kubernetes          ClusterIP     10.43.0.1    <none>        443/TCP          128m
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ nano deployment.yaml
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ nano k8s/deployment.yaml
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ kubectl apply -f k8s/
kubectl delete pods --all
deployment.apps/devops-app configured
service/devops-service unchanged
pod "devops-app-55b4986454-gn9ws" deleted from default namespace
pod "devops-app-55b4986454-gqkq2" deleted from default namespace
pod "devops-app-848bd76fb9-x/gw/" deleted from default namespace
^Cubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
devops-app-55b4986454-b2r7b   1/1     Running   0           17s
devops-app-55b4986454-gn9ws   1/1     Terminating   0           13m
devops-app-55b4986454-gqkq2   1/1     Terminating   0           13m
devops-app-55b4986454-gvtz2   1/1     Running   0           17s
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ kubectl get svc
NAME                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
devops-service      NodePort      10.43.19.120 <none>        80:30007/TCP     21m
kubernetes          ClusterIP     10.43.0.1    <none>        443/TCP          137m
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ kubectl delete deployment devops-app
deployment.apps "devops-app" deleted from default namespace
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ kubectl apply -f k8s/
deployment.apps/devops-app created
service/devops-service unchanged
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
devops-app-55b4986454-b2r7b   1/1     Terminating   0           3m27s
devops-app-55b4986454-bbrvn   1/1     Running        0           9s
devops-app-55b4986454-dcgjw   1/1     Running        0           9s
devops-app-55b4986454-gvtz2   1/1     Terminating   0           3m27s
ubuntu@ip-172-31-14-240:~/devops-project/devops-resume-project$
```

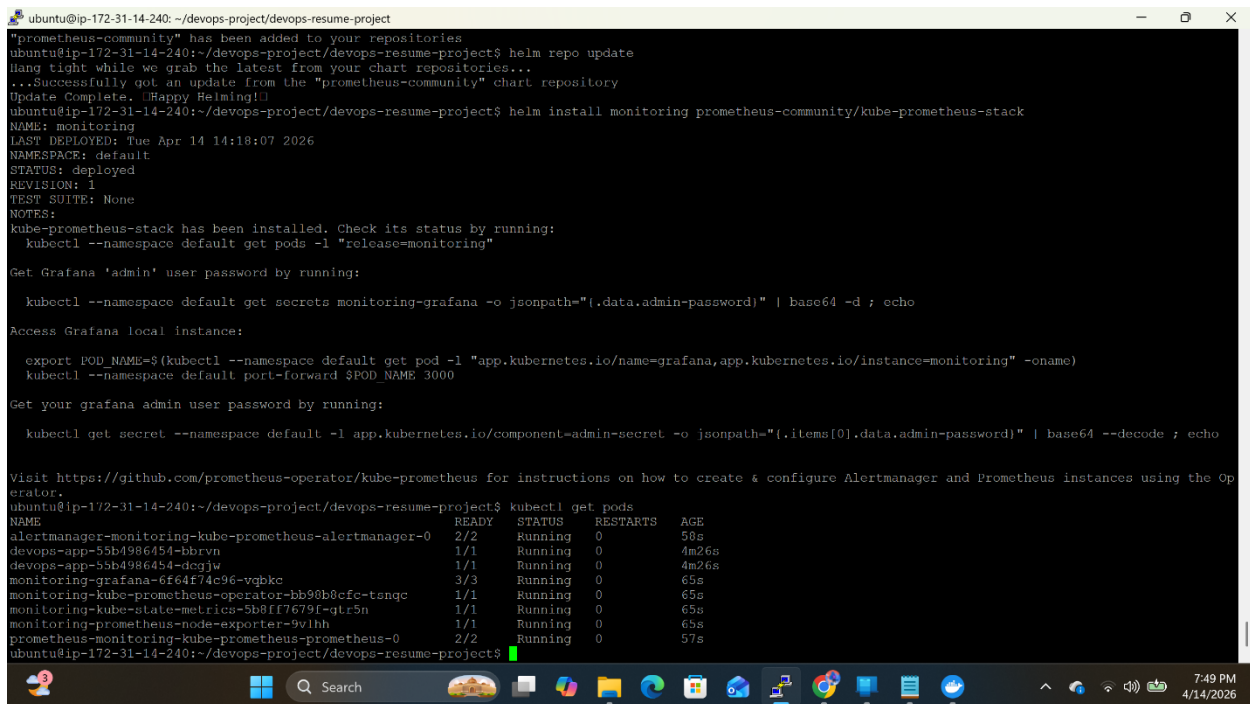
Output in Kubernetes container:



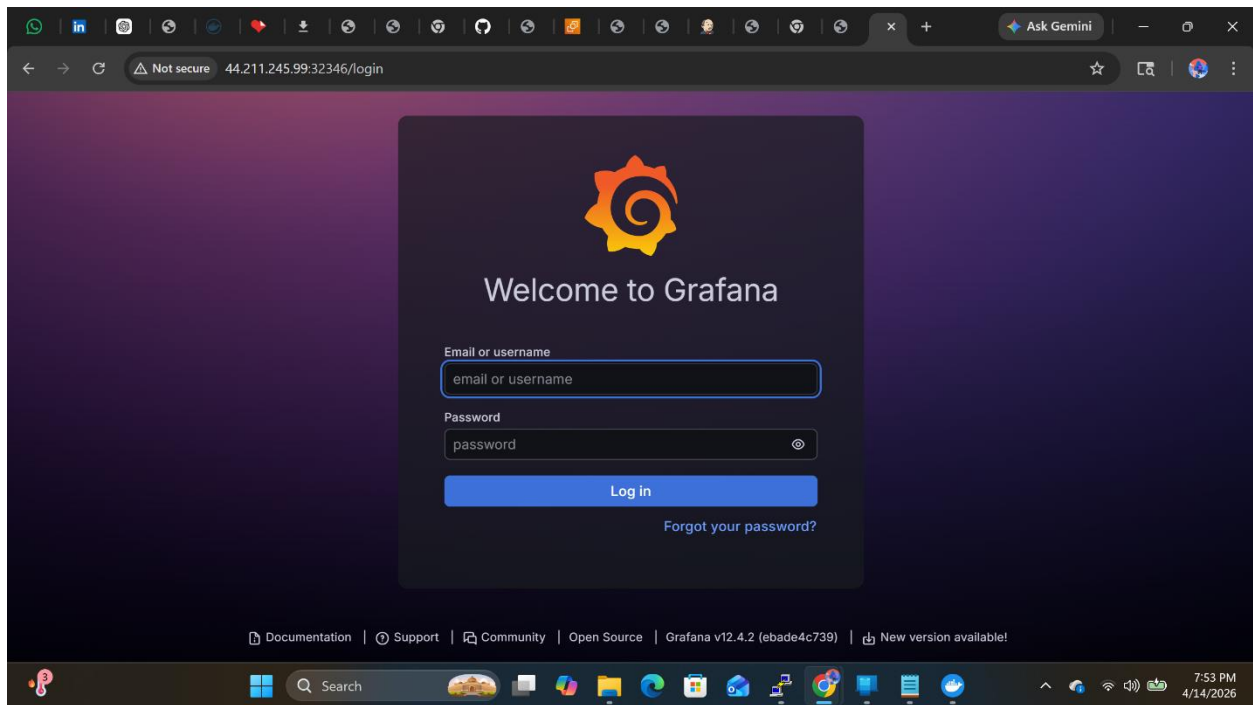
Jenkins pipeline success:



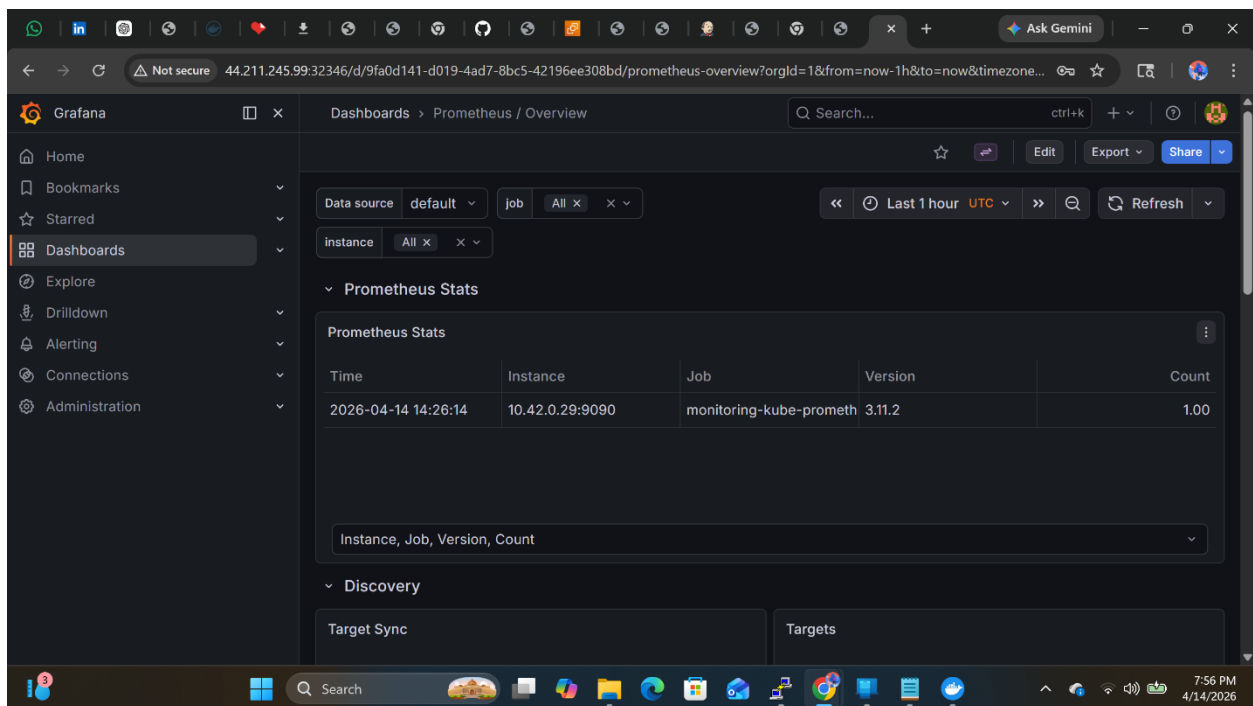
Prometheus installed and Grafana started:



Grafana login page:



Grafana Dashboard:



Grafana monitoring:

